

Project Design Phase-I

Solution Architecture

Date	1 March 2026
Team ID	699d5f82fdf409fdbd1bec66
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Solution Architecture:

Malaria diagnosis traditionally relies on manual microscopic examination of blood smear samples by trained pathologists, which is time-consuming, labor-intensive, and prone to human error. In regions with limited medical resources, the shortage of trained professionals further delays diagnosis and treatment.

To address this problem, the proposed solution uses Artificial Intelligence and Deep Learning techniques to automate the malaria detection process. Convolutional Neural Networks (CNN) are used to analyze microscopic blood smear images and classify them as Parasitized or Uninfected.

The system integrates deep learning models, image preprocessing techniques, and a web-based application to provide a reliable and scalable solution for malaria diagnosis. By leveraging technologies such as TensorFlow, Keras, and Flask, the system enables faster, more accurate, and accessible disease detection.

The malaria detection system is designed as a multi-layer architecture consisting of several interconnected components that work together to perform automated disease detection.

The system includes the following layers:

- **User Interface Layer** – A web-based interface that allows doctors or laboratory technicians to upload microscopic blood smear images for analysis.
- **Application Layer** – The backend server built using Flask handles user requests, processes uploaded images, and communicates with the machine learning model.
- **Data Processing Layer** – Image preprocessing techniques such as resizing, normalization, and augmentation prepare the input data for model prediction.
- **Machine Learning Layer** – Deep learning models including **Custom CNN**, **MobileNetV2**, and **EfficientNetB0** analyze the images and classify them into infected or uninfected categories.
- **Prediction and Visualization Layer** – The system returns the prediction results along with confidence scores, which are displayed to the user through the web interface.

This structured architecture helps stakeholders understand how the system processes data and delivers automated malaria diagnosis.

Key Features

- Automated malaria parasite detection using deep learning models
- Image preprocessing for improving prediction accuracy
- Real-time prediction through a web application
- Visualization of prediction results with confidence scores
- Scalable architecture for handling multiple image inputs

Development Phases

1. **Dataset Acquisition** – Collect malaria cell image dataset from Kaggle.
2. **Data Preprocessing** – Clean and prepare images through resizing, normalization, and augmentation.
3. **Model Development** – Train deep learning models including Custom CNN, MobileNetV2, and EfficientNetB0.
4. **Model Evaluation** – Evaluate model performance using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
5. **Application Development** – Build a web-based interface using Flask for user interaction.
6. **Deployment** – Deploy the trained model and application to enable real-time malaria detection.

Technical Specifications

- Programming Language: Python
- Deep Learning Framework: TensorFlow and Keras
- Web Framework: Flask
- Data Processing Libraries: NumPy, Pandas
- Visualization Tools: Matplotlib, Seaborn
- Machine Learning Metrics: Scikit-learn
- Image Processing: Pillow (PIL)

The trained model is exported as a **.h5 model file**, which is loaded by the Flask backend for inference. When a user uploads an image, the system preprocesses the input, runs it through the trained model, and returns the prediction results through the web interface.

This specification ensures that the solution can be easily managed, maintained, and extended in the future, enabling further improvements such as cloud deployment or integration with hospital information systems.

Solution Architecture Diagram:

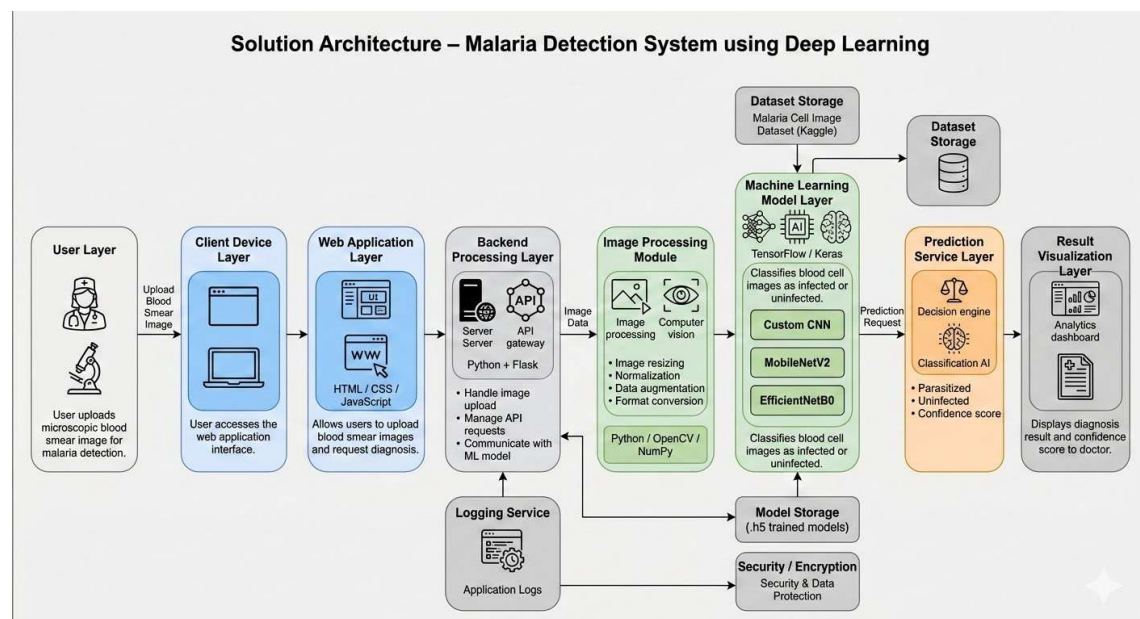


Figure 1: Solution Architecture of the Malaria Detection System using Deep Learning